Why plants transport sugars as sucrose and not as starch or glucose or fructose?

Glucose and Fructose are simple monosaccharides, whereas, Sucrose is a disaccharide composed of glucose and fructose. Starch is a polysaccharide of glucose. Sucrose and starch are more efficient in energy storage when compared to glucose and fructose, but starch is insoluble in water. So it cannot be transported via phloem and the next choice is sucrose, being water soluble and energy efficient, sucrose is chosen as the carrier of energy from leaves to different parts of the plant. Sucrose has low viscosity even at high concentrations and has no reducing ends which makes it inert than glucose or fructose.During photosynthesis, starch is synthesized and stored in the chloroplast stroma and sucrose is synthesized in the leaf cytosol from which it diffuses to the rest of the plant.

11.7.6 Phloem Unloading

From sieve elements sucrose is translocated into sink organs such as roots, tubers, flowers and fruits and this process is termed as **phloem unloading**. It consists of three steps:

1. ***Sieve element unloading***: Sucrose leave from sieve elements.
2. ***Short distance transport***: Movement of sucrose to sink cells.
3. ***Storage and metabolism***: The final step when sugars are stored or metabolized in sink cells.

11.7.7 Mechanism of Translocation

Several hypotheses have been proposed to explain the mechanism of translocation. Some of them are given below:

1. Diffusion hypothesis

As in diffusion process, this theory states the translocation of food from higher concentration (from the place of synthesis) to lower concentration (to the place of utilization) by the simple physical process. However, the theory was rejected because the speed of translocation is much higher than simple diffusion and translocation is a biological process which any poison can halt.

2. Activated diffusion theory

This theory was first proposed by **Mason** and **Maskell** (1936). According to this theory, the diffusion in sieve tube is accelerated either by activating the diffusing molecules or by reducing the protoplasmic resistance to their diffusion.

3. Electro-Osmotic theory

The theory of electro osmosis was proposed by **Fenson** (1957) and **Spanner** (1958). According to this, an electric-potential across the sieve plate causes the movement of water along with solutes. This theory fails to explain several problems concerning translocation.

4. Munch Mass Flow hypothesis

Mass flow theory was first proposed by **Munch** (1930) and elaborated by **Crafts** (1938). According to this hypothesis, organic substances or solutes move from the region of high osmotic pressure (from mesophyll) to the region of low osmotic pressure along the turgor pressure gradient. The principle involved in this hypothesis can be explained by a simple physical system as shown in figure 11.22.

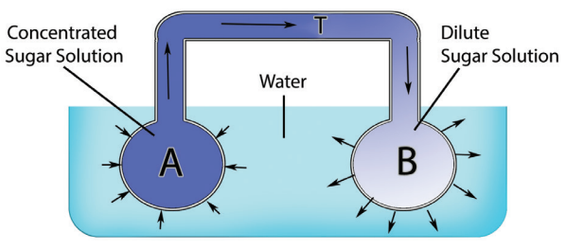


Figure 11.22: A model demonstrating the Mass flow hyphothesis

Two chambers "A" and "B" made up of semipermeable membranes are connected by tube "T" immersed in a reservoir of water. Chamber "A" contains highly concentrated sugar solution while chamber "B" contains dilute sugar solution. The following changes were observed in the system,

i. The high concentration sugar solution of chamber "A" is in a hypertonic state which

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